

ROS 2 Basics for Robotics

ROS 2

ROS 2 is a middleware framework used to build distributed robot applications. It provides communication, tools, libraries, and interfaces for robotic software.

Nodes

A node is a process that performs a specific function, such as reading a camera, controlling motors, or performing localization.

Topics

Topics use a publish-subscribe communication model. Publishers send messages and subscribers receive messages.

Services

Services provide request-response communication between a client and a server.

Actions

Actions support long-running tasks such as navigation and provide feedback while a goal is executing.